



VOICE-ACTUATED TRACKING SYSTEM FOR THE TWIN ROTOR PLATFORM

GEOMETRIC CONTROL | AI AGENT ORCHESTRATION | HUMAN-IN-THE-LOOP SAFETY

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A. INTRODUCTION

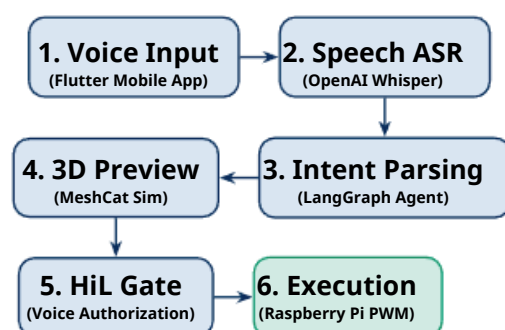
Operating aerodynamic platforms via manual workstations limits accessibility and rapid intervention. TwinTalk bridges human speech and complex aerospace hardware using an AI agent pipeline. By combining speech recognition, stateful orchestration, dynamic 3D simulation, and a mandatory Human-in-the-Loop authorization gate, it enables safe, hands-free trajectory tracking.

B. DEPLOYED SOFTWARE & HARDWARE

- **Mobile App:** Flutter/Dart (Android).
- **Speech Engine:** OpenAI Whisper API.
- **AI Agent:** LangGraph (Python).
- **Simulation:** MeshCat (Python/Three.js).
- **Control & Validation:** Raspberry Pi OS.
- **Twin Rotor MIMO System (TRMS):** Dual-rotor body (R1 / R2) on a pivot base.
- **Compute Unit:** Raspberry Pi 4 Model B (handling real-time hardware PWM generation).
- **Communication Module:** High-speed local Bluetooth / Wi-Fi connection.

C. METHODOLOGY & SAFETY GATE

The pipeline forces operator intent through rigorous verification before hardware dispatch.

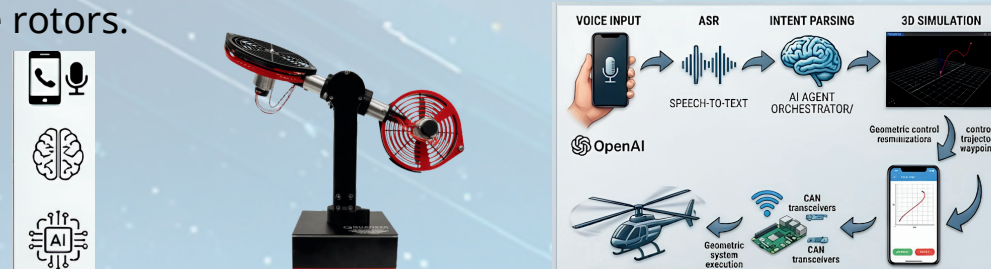


D. FINAL SYSTEM INTEGRATION SUMMARY

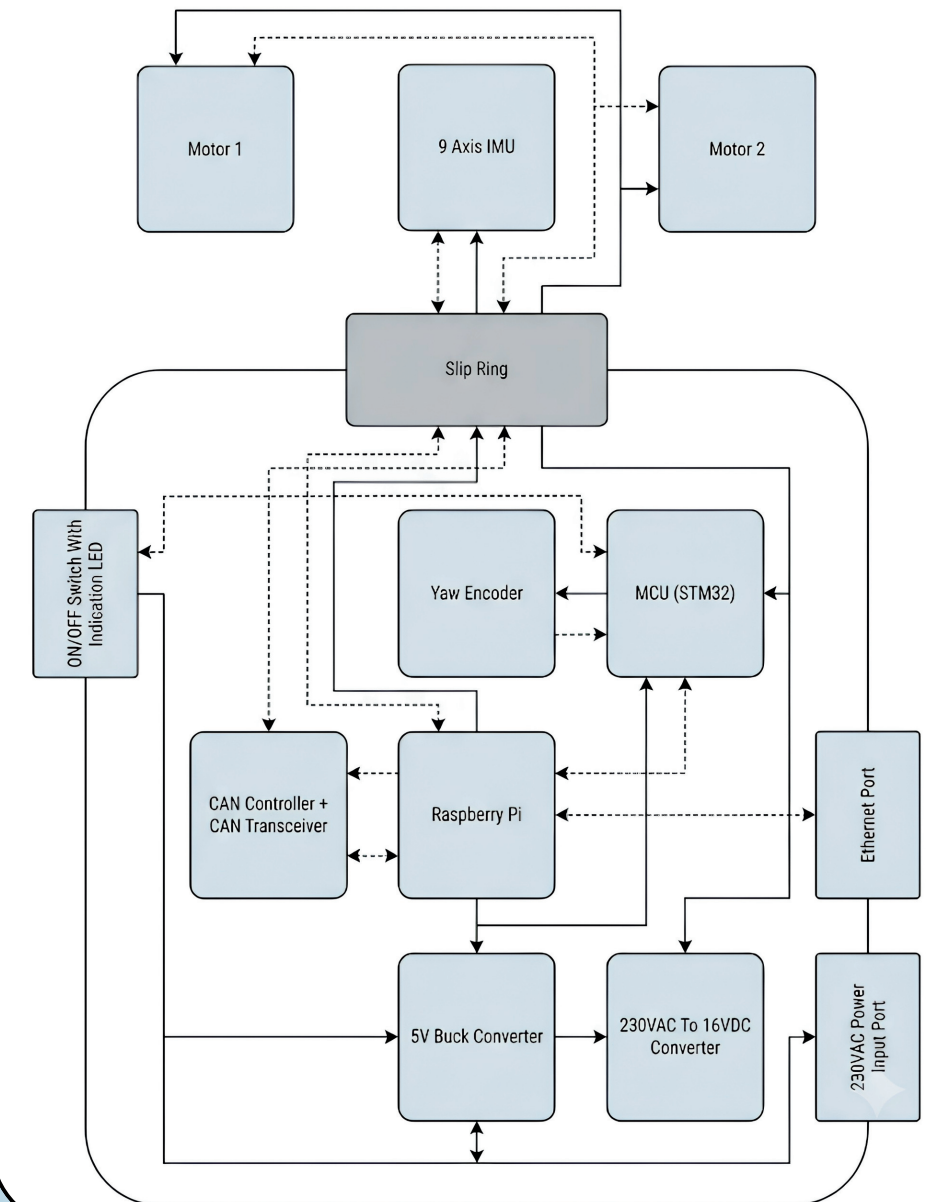
The TwinTalk system successfully bridges natural human speech and precision aerodynamic hardware, replacing restrictive manual inputs with a hands-free interface.

Core Architecture & Work flow:

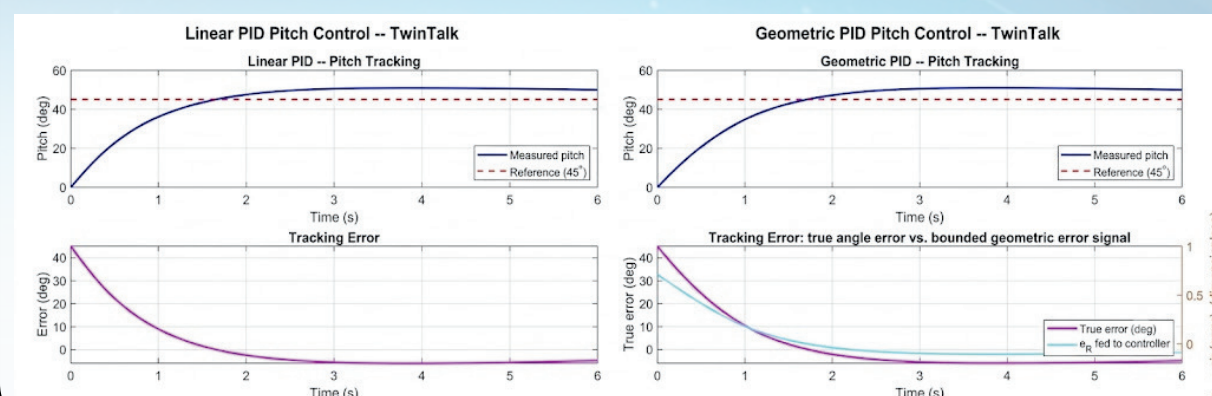
- **Voice Input:** The operator issues a natural language trajectory command through a custom mobile application.
- **AI Orchestrator:** The mobile app streams the audio to a central laptop, where OpenAI Whisper translates the speech, and a multi-agent LangGraph system parses the operator's intent.
- **End-to-End Pipeline:** Successfully integrated a six-step software pipeline from Voice → AI → Simulation → HiL → Hardware Dispatch.
- **Safety Evaluation:** The AI Orchestrator evaluates the command. If the trajectory is complex or high-risk, it routes the command to an offline MeshCat 3D simulation environment.
- **Execution:** Only when the user provides explicit voice verification ("Authorize" / "Execute") does the laptop command the Raspberry Pi 4 Model B to output precise hardware PWM signals to the rotors.



F. ELECTRICAL DIAGRAM



E. LINEAR AND GEOMETRIC PIDs



G. BASIC COMPONENTS

